PRANA-TWIN End-to-End System Architecture

Multi-Stream Data Fusion, Physiological ODEs, Machine Learning Predictors, and Doctor Dashboard

Layer 1: Multi-Stream Data Ingestion

• Static EHR: Demographics, HbA1c (8.6%), eGFR, Lipids, Meds
• Dynamic IoT: 5-min CGM Glucose, PPG Pulse, HRV (RMSSD), Steps
• Indian Dietary Profile: High-GI staple meal carbs (Rice, Parathas)

Layer 2: Hybrid Digital Twin Core

• Physiological Compartment ODEs (Bergman Minimal Model)
• Multimodal Fusion Engine & Temporal Derivative Extractor (dG/dt)
• Gradient Boosted Regressor (120-min forward trajectory)
• Adverse Event Classifiers (Spike >220 mg/dL & Hypo <70 mg/dL)
• Counterfactual 'What-If' Simulation Sandbox Engine

Layer 3: Clinical Decision Support UI

• Patient Triage Selector (Critical, High Risk, Moderate, Stable)
• Interactive Virtual Human Organ Twin (Pancreas, Heart, Kidney, Liver)
• 2-Hour Forward Horizon Graph with 90% Confidence Bands
• Counterfactual Sandbox (Sliders for Carbs, Walk, Meds)
• One-Click RSSDI / ICMR Clinical Audit & Ambulatory Glucose Report